SELT Text Classification Project Final Presentation

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Note: In accordance with confidentiality requirements, the actual data and its resulting insights will be redacted.

Project Overview

What was the project about?

Project Objectives and Scope

In this project, we will leverage pre-trained transformer models and build a scalable solution for continuous feedback analysis. Given the question format in the actual survey, sentiment analysis becomes less meaningful. Therefore, we will focus on developing **two classification objectives**:

Data

1. Synthetic student survey data generated by ClaudeAI.
2. SELT Survey Response 2024.

Technical Approach

1. Data pre-processing.
2. Data exploration.
3. Model setup and training.
4. Performance evaluation and optimisation.

Expected Outcomes

Appropriateness Classifier

- Appropriate
- Inappropriate

Categorization Classifier

- Assignment & Quiz
- Course Materials & Structure
- Teaching & Delivery

Student Survey Synthetic Data

Generating data for model training.

Generating Synthetic Data

The process of generating synthetic data still reflects the format of the actual student survey to replicate the real use case and support future implementation on the actual data. The synthetic data were generated using **ClaudeAI** with two accounts, each account focusing on one question.

The prompt

Generate 100 rows of synthetic data for student survey responses. The survey question is "Which aspects of this course need improvement?" . Some students provide a comprehensive and structured response, while others may write short or uninformative responses just to mention the high level improvements that need to be addressed.

I want the structure of the data to have 4 columns, since it will be used for text classification. Include 2 columns related to the response: course name and student response and 2 columns of labels: inappropriate/appropriate and category.

Generate responses with a variety of length (around 200 words) and can you help me to only include course name from Uni Adelaide (not the course code):

https://www.adelaide.edu.au/course-outlines/ug/set/

https://www.adelaide.edu.au/course-outlines/ug/able/

https://www.adelaide.edu.au/course-outlines/ug/hms/

The inappropriate/appropriate label is used to catch responses that are mentioning sensitive topic, rude, racist, or not respectful to the teaching staffs.

The category will be separated into 3 classes and below are the details for each class:

1. Course materials & structure: the content or materials used in the course, and the structure of the course related to materials access, weighting distribution, and schedule.
2. Teaching & delivery: delivery of lecture session (in person or videos), practical, and workshop or related to the teaching staff.
3. Assignment & quiz: assignment and quiz tasks, instruction clarity, and due date.

The result

1	course_name	student_response	inappropriate_appropria	category
2	Introduction to Computer Science	the lecture recordings are often unclear and the audio quality needs significant improvement. also, the practical sessions feel rushed and we don't get enough time to complete the exercises properly. the programming assignments are challenging which is good, but the debugging workshops need more structured guidance. when we encounter errors, the tutors often just tell us to figure it out ourselves rather than providing systematic approaches to problem-solving. i think having more step-by-step debugging tutorials and maybe peer programming sessions would really help students develop better coding practices and confidence in tackling complex problems independently.	appropriate	teaching & delivery
3	Mathematical Methods	this course sucks and prof smith is the worst teacher ever. complete waste of time.	inappropriate	teaching & delivery
4	Physics for Engineers	the assignment instructions are very ambiguous and the due dates are too close together, making it difficult to manage workload effectively. the theoretical concepts are explained well in lectures, but the practical applications in problem-solving sessions need more variety. many of the examples used are quite similar and don't prepare us adequately for the diverse range of questions that appear in assessments. i would suggest incorporating more real-world engineering scenarios that demonstrate how these physics principles actually apply in professional practice. additionally, the laboratory equipment is sometimes malfunctioning which delays our experimental work and affects our ability to collect accurate data for our reports.	appropriate	assignment & quiz
5	General Chemistry	the lab manual is outdated and some experiments don't work as described. would benefit from updated materials and better equipment.	appropriate	course materials & structure
6	Introduction to Biology	good course overall.	appropriate	teaching & delivery
7	Principles of Economics	the textbook is extremely expensive and doesn't align well with the lecture content. more affordable alternatives should be considered. the economic models we study are interesting but i feel like we need more discussion about their limitations and assumptions. sometimes the mathematical components are presented without sufficient explanation of what they mean in practical economic terms. it would be helpful to have more case studies showing how these theories apply to current australian economic situations rather than just focusing on abstract examples. the tutorial discussions are valuable but could be enhanced with more structured debate formats that encourage different perspectives on economic issues.	appropriate	course materials & structure
8	Medieval History	the essay topics are interesting but the marking criteria isn't clear enough. i'd appreciate more detailed rubrics and examples of high-quality work.	appropriate	assignment & quiz
9	Introduction to Psychology	lectures are boring as hell and the tutor doesn't know what she's talking about.	inappropriate	teaching & delivery
10	Philosophy and Ethics	the course structure is well-organized, but the weekly quizzes feel repetitive and don't test understanding of key concepts effectively.	appropriate	assignment & quiz

Number of data: 675 training rows (338 rows each) + 100 test rows (50 rows each, generated with a different prompt to test model on new, unseen data)

Generating Synthetic Data

The **survey questions** that the responses correlate to:

1. What are the best aspects of this course?
2. Which aspects of this course need improvement?

The labels generated by the AI are:

1. **Appropriate / Inappropriate**
2. **Category**: course materials & structure, teaching & delivery, assignment & quiz.

The generated synthetic data was manually reviewed, and additional data generation was performed during the model training phase to test the model on unseen data.

We also manually created **two new labels** for the categorisation to explore the model performance with updated category labels. These labels mark responses that contain more than one category as '**mixed**' and apply a **multi-label** approach by explicitly labelling responses with multiple categories.

Final synthetic data

1	course_name	student_response	inappropriate_appropria	category	category_mixed	all_labels
2	Introduction to Computer Science	the lecture recordings are often unclear and the audio quality needs significant improvement. also, the practical sessions feel rushed and we don't get enough time to complete the exercises properly. the programming assignments are challenging which is good, but the debugging workshops need more structured guidance. when we encounter errors, the tutors often just tell us to figure it out ourselves rather than providing systematic approaches to problem-solving. i think having more step-by-step debugging tutorials and maybe peer programming sessions would really help students develop better coding practices and confidence in tackling complex problems independently.	appropriate	teaching & delivery	teaching & delivery	['teaching & delivery']
3	Mathematical Methods	this course sucks and prof smith is the worst teacher ever. complete waste of time.	inappropriate	teaching & delivery	mixed	['teaching & delivery', 'course materials & structure']
4	Physics for Engineers	the assignment instructions are very ambiguous and the due dates are too close together, making it difficult to manage workload effectively. the theoretical concepts are explained well in lectures, but the practical applications in problem-solving sessions need more variety. many of the examples used are quite similar and don't prepare us adequately for the diverse range of questions that appear in assessments. i would suggest incorporating more real-world engineering scenarios that demonstrate how these physics principles actually apply in professional practice. additionally, the laboratory equipment is sometimes malfunctioning which delays our experimental work and affects our ability to collect accurate data for our reports.	appropriate	assignment & quiz	mixed	['assignment & quiz', 'course materials & structure']
5	General Chemistry	the lab manual is outdated and some experiments don't work as described. would benefit from updated materials and better equipment.	appropriate	course materials & structure	course materials & structure	['course materials & structure']
6	Introduction to Biology	good course overall.	appropriate	teaching & delivery	course materials & structure	['course materials & structure']
7	Principles of Economics	the textbook is extremely expensive and doesn't align well with the lecture content. more affordable alternatives should be considered. the economic models we study are interesting but i feel like we need more discussion about their limitations and assumptions. sometimes the mathematical components are presented without sufficient explanation of what they mean in practical economic terms. it would be helpful to have more case studies showing how these theories apply to current australian economic situations rather than just focusing on abstract examples. the tutorial discussions are valuable but could be enhanced with more structured debate formats that encourage different perspectives on economic issues.	appropriate	course materials & structure	course materials & structure	['course materials & structure']
8	Medieval History	the essay topics are interesting but the marking criteria isn't clear enough. i'd appreciate more detailed rubrics and examples of high-quality work.	appropriate	assignment & quiz	assignment & quiz	['assignment & quiz']

Data Pre-processing

Prepares data for optimal model training.

Pre-processing Step

We **removed response with fewer than 10 characters**, as they become noisy text and lack semantic meaning, which can confuse the model training.

There are not many pre-processing steps required for the survey text, as the format is plain text. Two steps were performed to standardize the survey text:

1. **Keeping only alphanumeric characters** (a-z, A-Z, 0-9) to remove unnecessary icons, emojis, or other characters.
2. **Lowercasing** to standardize the text.

For the labels, we also perform **lowercasing** and **sorting** to standardize the format, as they were manually labeled.

During the model training, we also experimented with **removing stopwords** by excluding common words such as "the", "is", "and". However, this resulted in lower performance, as the language model performs better with the full text to better understand and capture the semantic meaning within the response.

Noisy text

	course_name	question	student_response
335	Abstract Algebra	What are the best aspects of this course?	bad
343	Number Theory	What are the best aspects of this course?	nothing
360	Financial Accounting	What are the best aspects of this course?	Nice.
377	International Relations	What are the best aspects of this course?	Good.
390	Operations Research	What are the best aspects of this course?	Fine.

Before cleaning

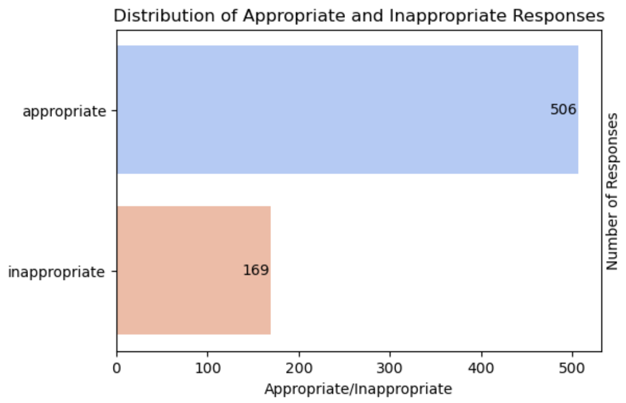
After cleaning

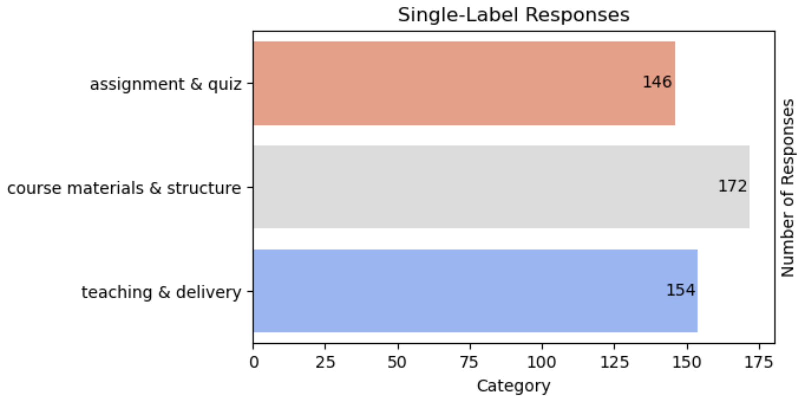
	student_response	student_response_preprocessed
0	Good balance of theory and practice.	good balance of theory and practice
1	The midterm exam had several errors and unclear questions that confused many students.	the midterm exam had several errors and unclear questions that confused many students
2	Course content is relevant but delivery could be more engaging with interactive elements.	course content is relevant but delivery could be more engaging with interactive elements
3	Assignment feedback takes too long to receive, sometimes weeks after submission.	assignment feedback takes too long to receive sometimes weeks after submission
4	The professor acts like they're doing us a favor by teaching and has a terrible attitude towards student questions. Total waste of money.	the professor acts like theyre doing us a favor by teaching and has a terrible attitude towards student questions total waste of money
5	Excellent problem sets that challenge understanding and application of concepts covered in lectures.	excellent problem sets that challenge understanding and application of concepts covered in lectures
6	The course covers essential topics like statistics and Python programming, but there are issues across multiple areas. The textbook is outdated with Python 2 examples while we're using Python 3, making it confusing to follow along. Lecture delivery is inconsistent - some sessions are engaging while others feel rushed. Assignment deadlines often overlap with other courses, and the grading criteria aren't always clear.	the course covers essential topics like statistics and python programming but there are issues across multiple areas the textbook is outdated with python 2 examples while were using python 3 making it confusing to follow along lecture delivery is inconsistent some sessions are engaging while others feel rushed assignment deadlines often overlap with other courses and the grading criteria arent always clear

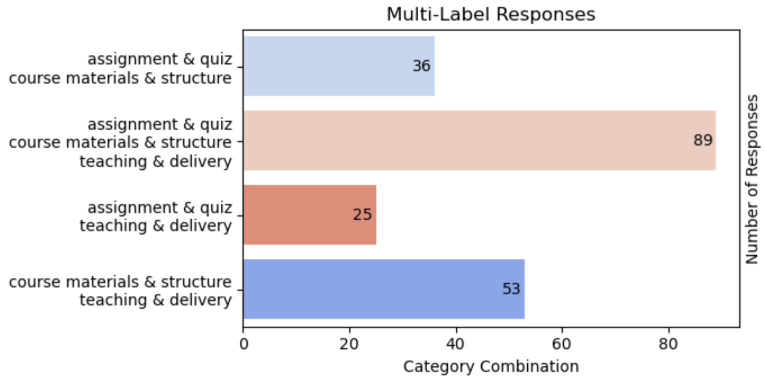
Data Exploration

Explore data through visualisation.

Initial Synthetic Data Distribution





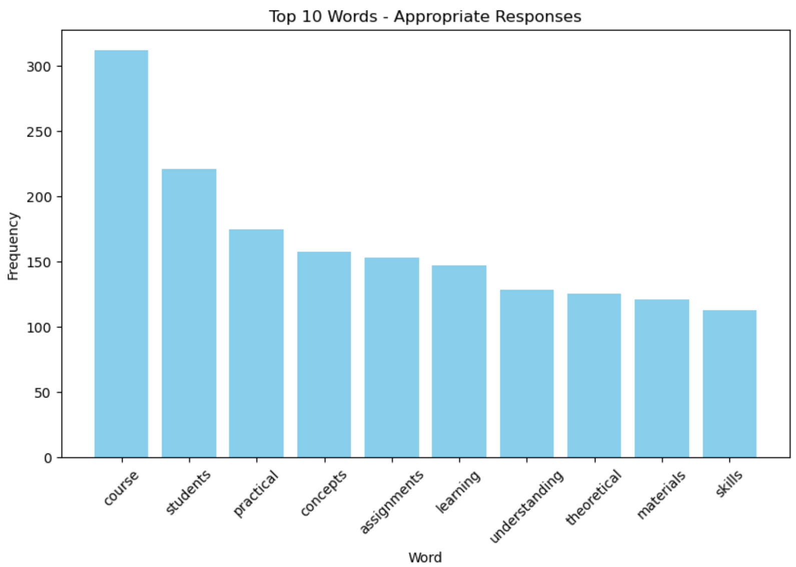


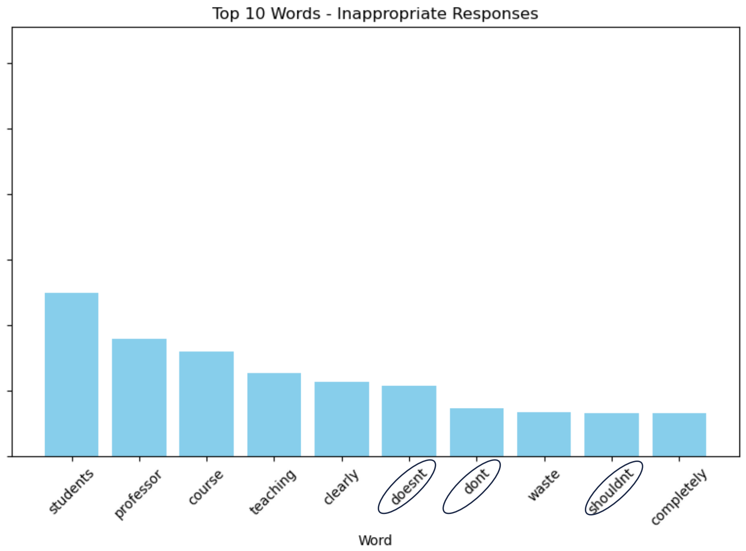
- **Significant imbalance between appropriate and inappropriate labels**, with inappropriate samples only one-third of the appropriate data.
- This reflects the actual response where appropriate responses are more common.

- We have a less noticeable imbalance between single-label classes.
- There are **more single-label samples compared to multi-label samples.**

- The portion of multi-label response still has **a sufficient proportion to be used for training the multi-label classifier.**

Most Common Words





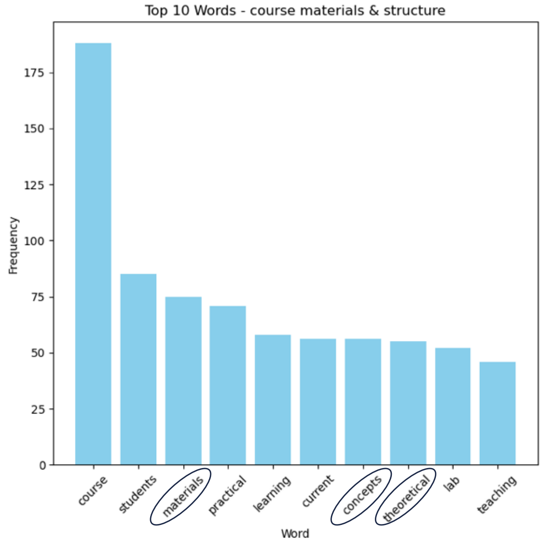
Appropriate Response

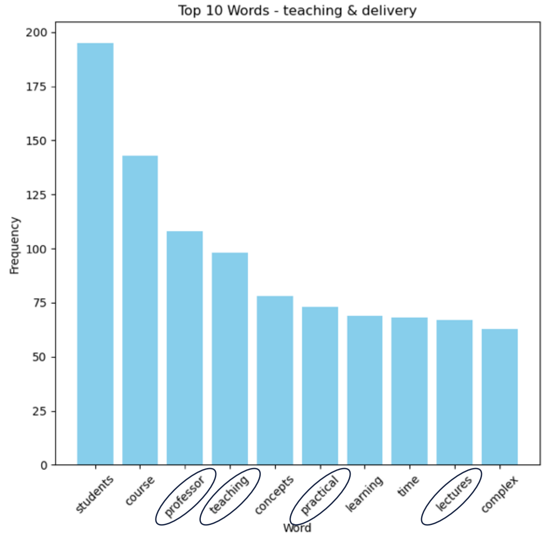
The steep decline in word frequency indicates that **appropriate feedback is highly concentrated around specific keywords** such as "course," "students," and "practical." This indicates that most appropriate responses focus on similar themes related to course structure and student experience.

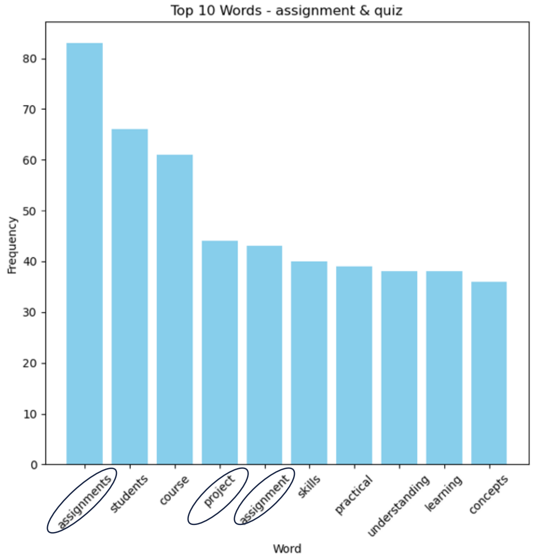
Inappropriate Response

The flatter distribution indicates **more diverse and scattered keywords across inappropriate responses**. Keywords such as "don't," "doesn't," and "shouldn't" appear among the top terms, showing that most inappropriate responses express negative sentiment that written in an informal manner.

Most Common Words







Course materials & structure

Definition: The content or materials used in the course, as well as the structure of the course related to materials access, weighting distribution, and schedule.

The dominant keywords in the data reflect this definition, such as “materials,” “concepts,” and “theoretical.”

Teaching & delivery

Definition: Delivery of lecture session (in person or videos), practical, workshops, or related to the teaching staff.

The dominant keywords in the data reflect this definition, such as “professor,” “teaching,” and “lectures.”

Assignment & quiz

Definition: Assignment and quiz tasks, clarity of instructions, and due dates.

The dominant keywords in the data reflect this definition, such as “assignments” and “project.”

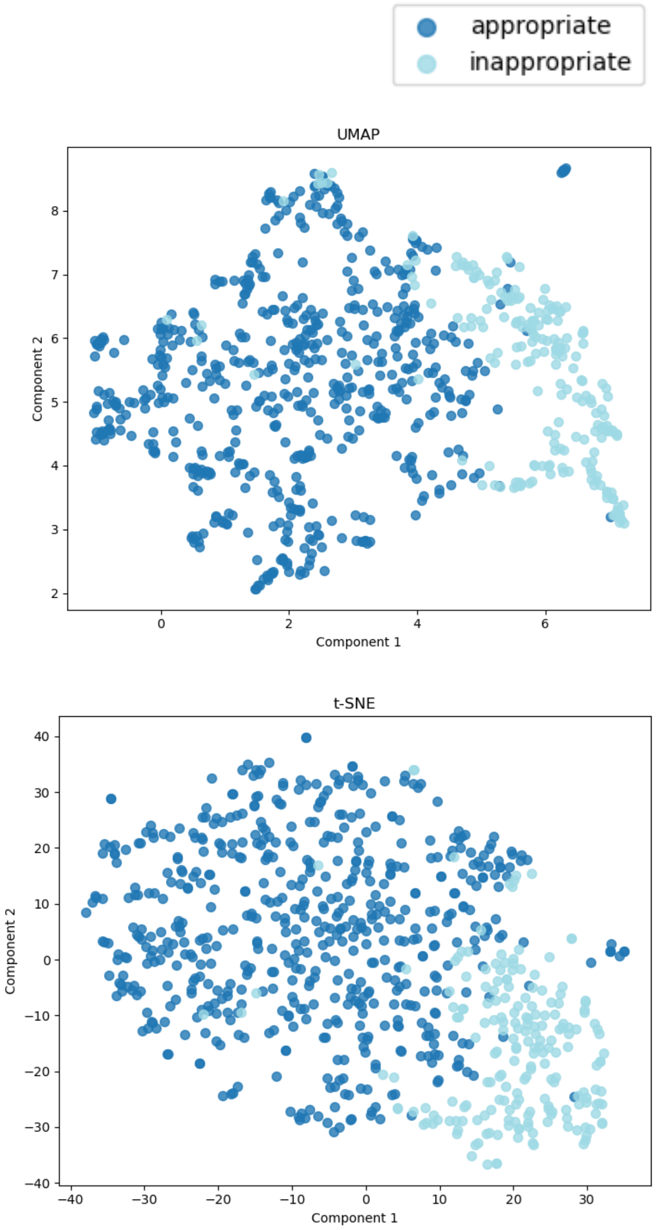
Label Quality Diagnostic

Visualization Methodology

- UMAP and t-SNE are visualization techniques that transform text into vector representation and project the high-dimensional vector into 2D.
- UMAP focuses on preserving both local and global data relationship, while t-SNE focuses on showing local groups. We use both methods to strengthens confidence in the findings.

Key Insights

- Both visualizations indicate **moderate separation between appropriate and inappropriate** responses.
- The overlap indicates that the distinction between the two relies on subtle wording rather than obvious differences. However, the separation shows potential for a model to learn distinguish these subtle differences, especially in the overlapping areas.



Label Quality Diagnostic

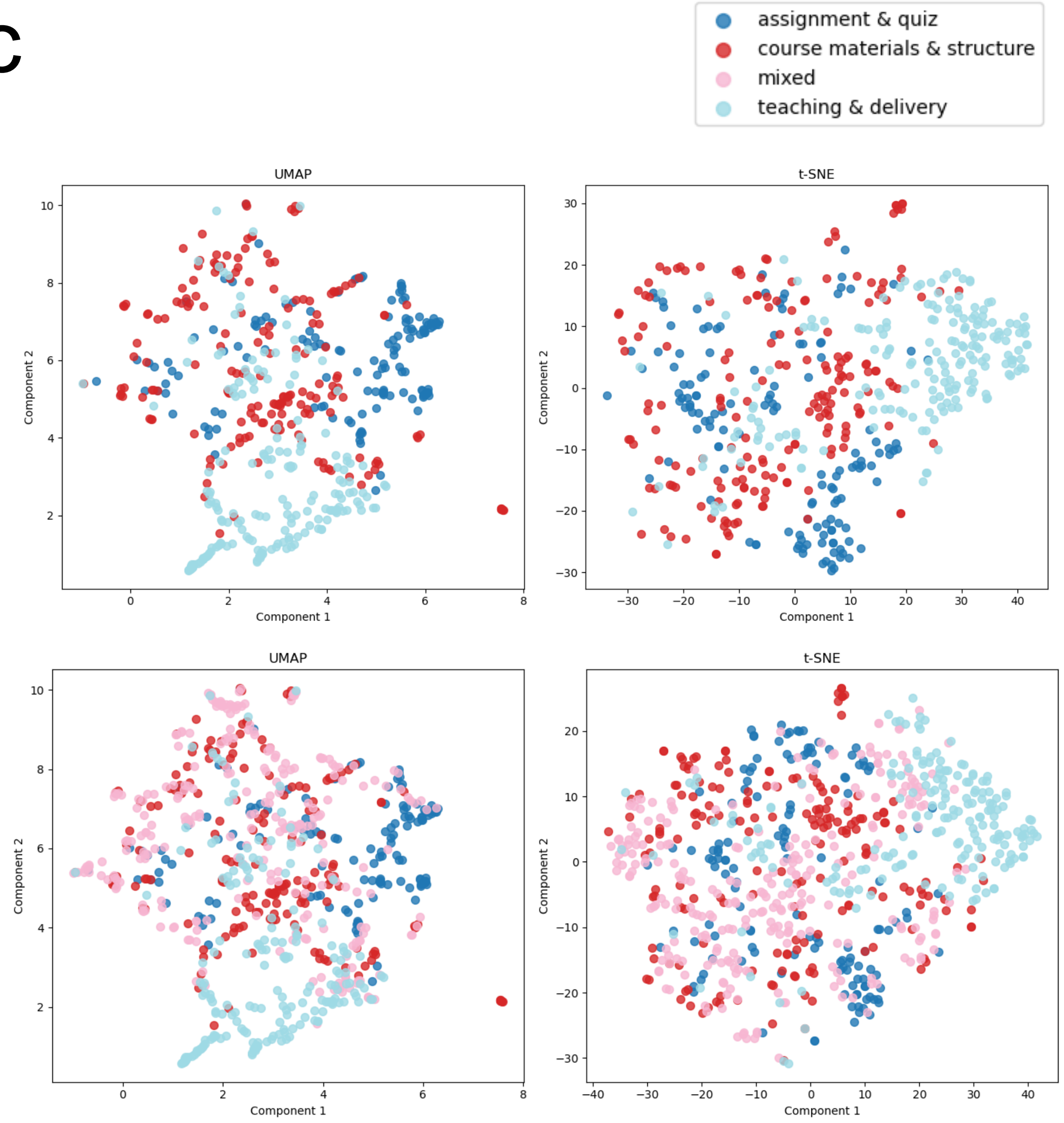
Single-label response

- We can see **clearer clustering patterns in responses with single-label**, with “teaching & delivery” forming a more distinct cluster. The patters are **more scattered but shows some visible clustering** within each category.

Multi-label response

- We also assessed responses containing multiple categories (labeled as ‘mixed’). The **multi-label group is scattered throughout the entire visualization area**, representing the overlapping content nature of this group.
- Clear separation is visible when the “mixed” group is removed. This indicates that **multi-label responses add complexity to the classification task** that become our main focus since many of the actual responses exhibit this behavior.

- Overall, moderate separation suggests that while categories have distinguishable patterns, there's substantial linguistic overlap between categories.



Methodology: Transformer-based Model

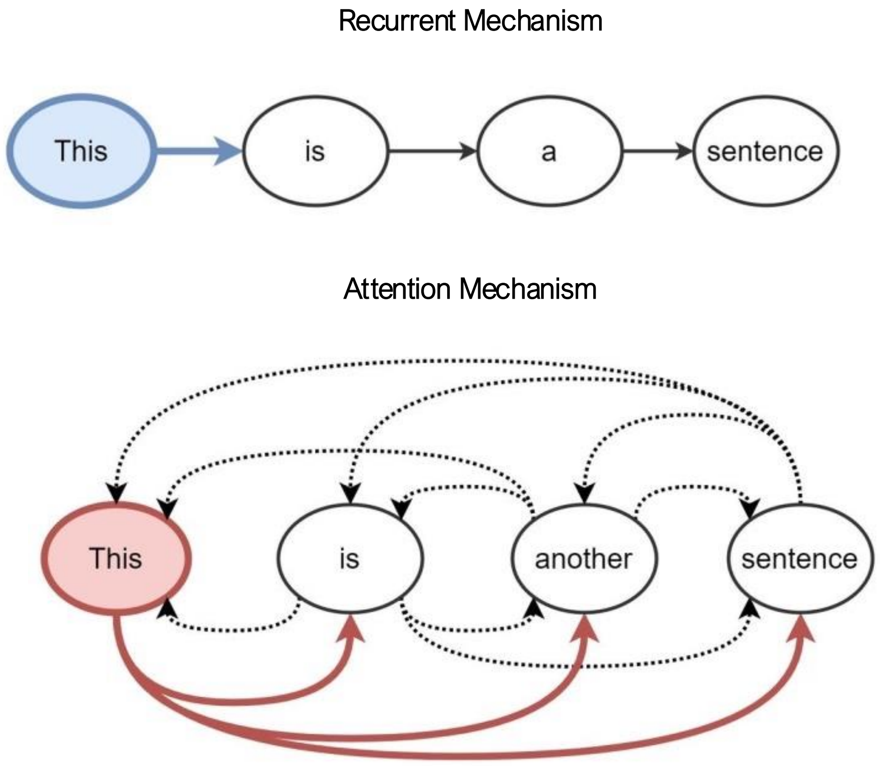
Understanding the concept behind the model.

Transformers

A transformer is a revolutionary neural network architecture introduced in “Attention is All You Need” in 2017. It leverages the **attention-mechanism** that eliminated the need for recurrent structures and allowing the model to **capture the meaning of each part of a sentence** without losing context in long texts (a limitation in recurrent networks).

Key Innovations:

- **Self-Attention**: Allows the model to focus on relevant parts of an input sequence.
- **Parallel Processing**: Enables simultaneous processing of all tokens in a sequence.
- **Position Encoding**: Maintains the understanding of word order without relying on recurrence.

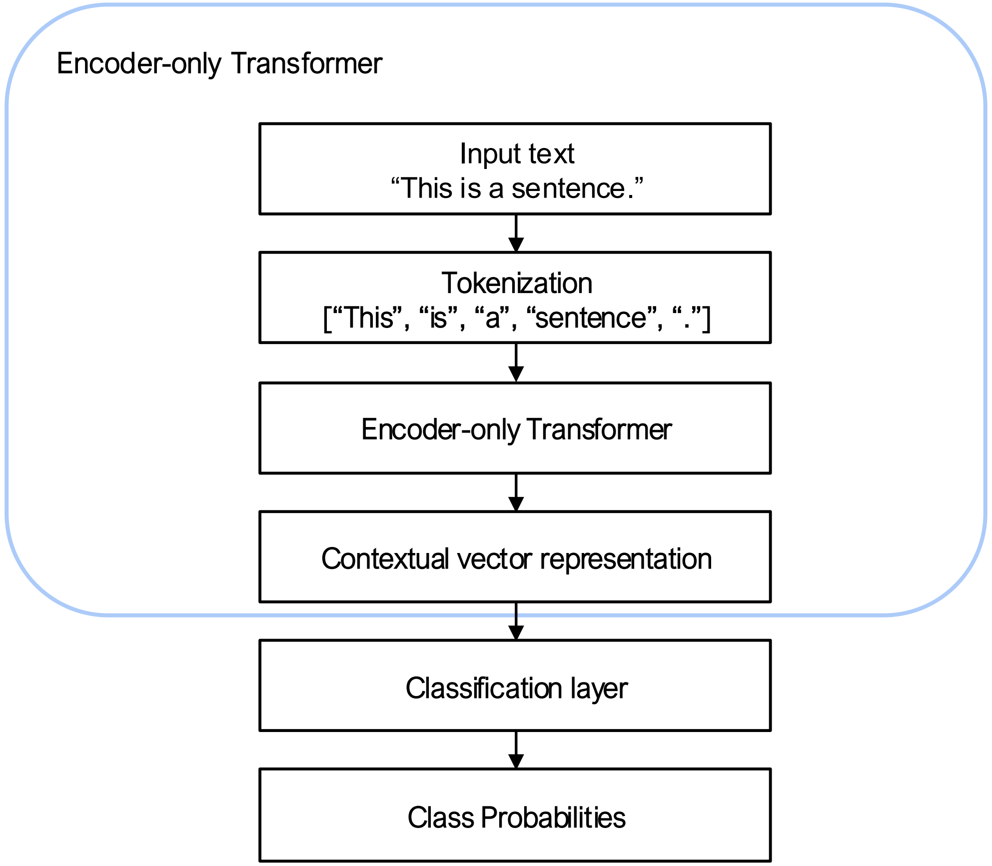


Encoder-Only Transformer

In this project, we experimented with DeBERTa and RoBERTa. Both are encoder-transformers, meaning they transform text into vector embeddings that are suitable for classification tasks.

Transformer Components

- **Encoder:** Converts an input sequence into a contextual vector representation that captures the meaning and relationships within the input.
- **Decoder:** Generates an output sequence using the encoder's representation to predict each output element iteratively.



DeBERTa and RoBERTa

Both DeBERTa and RoBERTa are advanced transformers, pre-trained on large corpora, and have demonstrated strong and significant performance in classification tasks.

RoBERTa (Robustly Optimised BERT)

- Improved version of BERT.
- Trains on larger datasets with longer sequences that allow better understanding of context within words.
- Performs well across different domains and datasets.

DeBERTA (Decoding-Enhanced BERT)

- Improved version of RoBERTa.
- Better contextual understanding through disentangled attention mechanism.
- State-of-the-art performance on many NLP benchmarks.

Advantages

- Rich language understanding
- Efficient training
- Superior performance

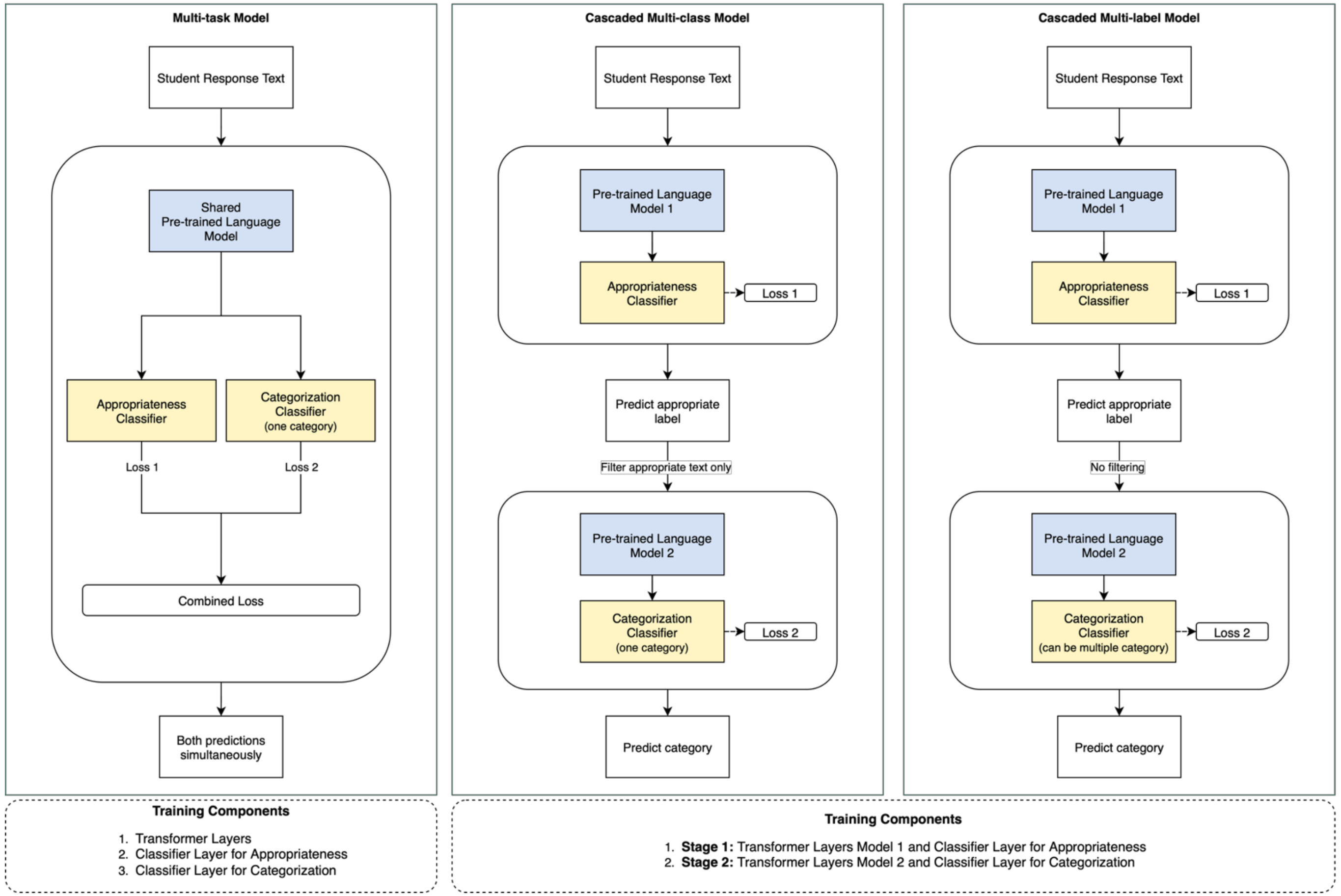
Implementation Strategy

- Use pre-trained DeBERTa and RoBERTa as base models.
- Add classification heads for our specific tasks.
- Fine-tune on the generated synthetic data.

Model Training & Evaluation

Two classification tasks, one model or two models?

Model Architecture



Multi-Task Model	Cascaded Multi-Class Model	Cascaded Multi-Label Model
Use **one single language model** to handle both tasks together.	Use **two separate language models**, one for each task.	Use **two separate language models**, one for each task.
Parallel training (both predictions happen at the same time).	**Sequential training** (one prediction at a time).	**Sequential training** (one prediction at a time).
Assigns **one category**.	Assigns **one category.**	Assigns **multiple categories**.
Efficient training, especially when tasks are related.	**Clear task separation** and focused learning.	**Clear task separation** and focused learning.
If one task dominates, the other can struggle to optimize.	Less efficient, but useful when tasks are not closely related.	Less efficient, but useful when tasks are not closely related.
Workflow Outline		
Learn the appropriateness and categorization tasks at the same time.	Learn the appropriateness task → Filter for appropriate only → Learn the categorization task.	Learn the appropriateness task → No filtering → Learn the categorization task.

Training – Appropriateness Classifier

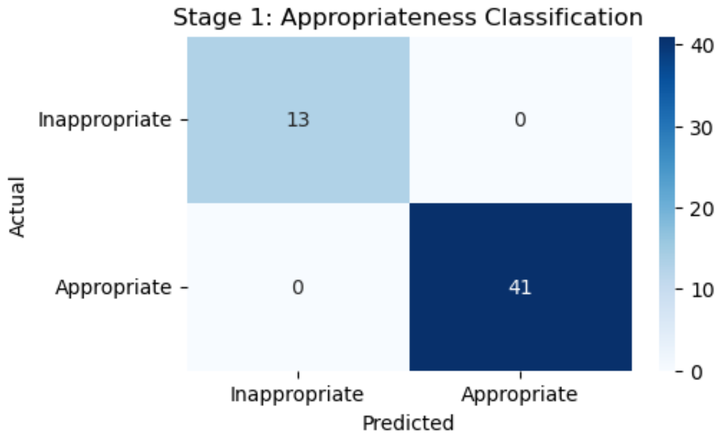
Model		Accuracy	Precision	Recall	F1-score
Multi-Task	DeBERTa	1.000	1.000	1.000	1.000
	RoBERTa	1.000	1.000	1.000	1.000
Cascaded Binary Classification	DeBERTa	1.000	1.000	1.000	1.000
	RoBERTa	1.000	1.000	1.000	1.000

Testing – Appropriateness Classifier

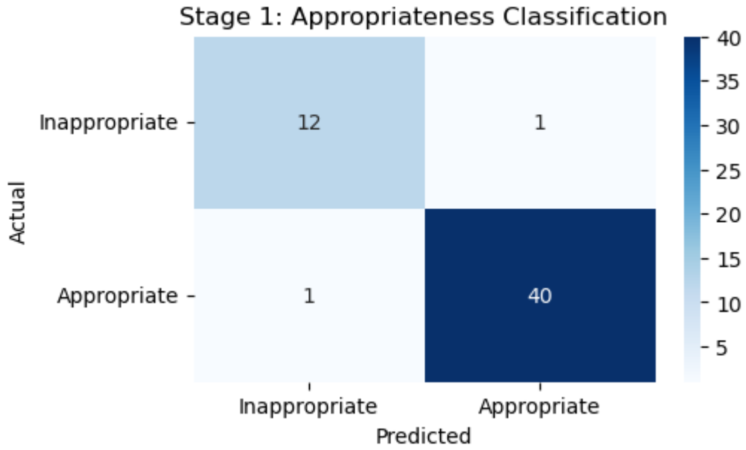
Model		Accuracy	Precision	Recall	F1-score
Multi-Task	DeBERTa	1.000	1.000	1.000	1.000
	RoBERTa	1.000	1.000	1.000	1.000
Cascaded Binary Classification	DeBERTa	0.933	0.935	0.933	0.931
	RoBERTa	0.702	0.702	1.000	0.825

Visualisation – Appropriateness Classifier

Multi-Task



Cascaded Binary Classification



The appropriateness classifier **performs well across all both model architectures**. In terms of architecture, the multi-task model is more lightweight since it uses only one encoder and a simple output layer without additional hidden layers. This indicates that the appropriateness task can be handled effectively by a simpler design. However, model selection will depend on the performance of the categorization classifier.

Training – Categorisation Classifier

Model		Accuracy	Macro F1-score
Multi-Task	DeBERTa	0.701	0.708
	RoBERTa	0.707	0.709
Cascaded Multi-Class	DeBERTa	0.793	0.782
	RoBERTa	0.463	0.315
Cascaded Multi-Label	DeBERTa	0.815	0.912
	RoBERTa	0.827	0.941

Since we found that the **cascaded multi-label approach outperforms the other methods**, we selected this approach as the most suitable architecture for our dataset and classification task.

Testing – Categorisation Classifier

Model		Accuracy	Macro F1-score
Cascaded Multi-Label	DeBERTa	0.587	0.809
	RoBERTa	0.606	0.818

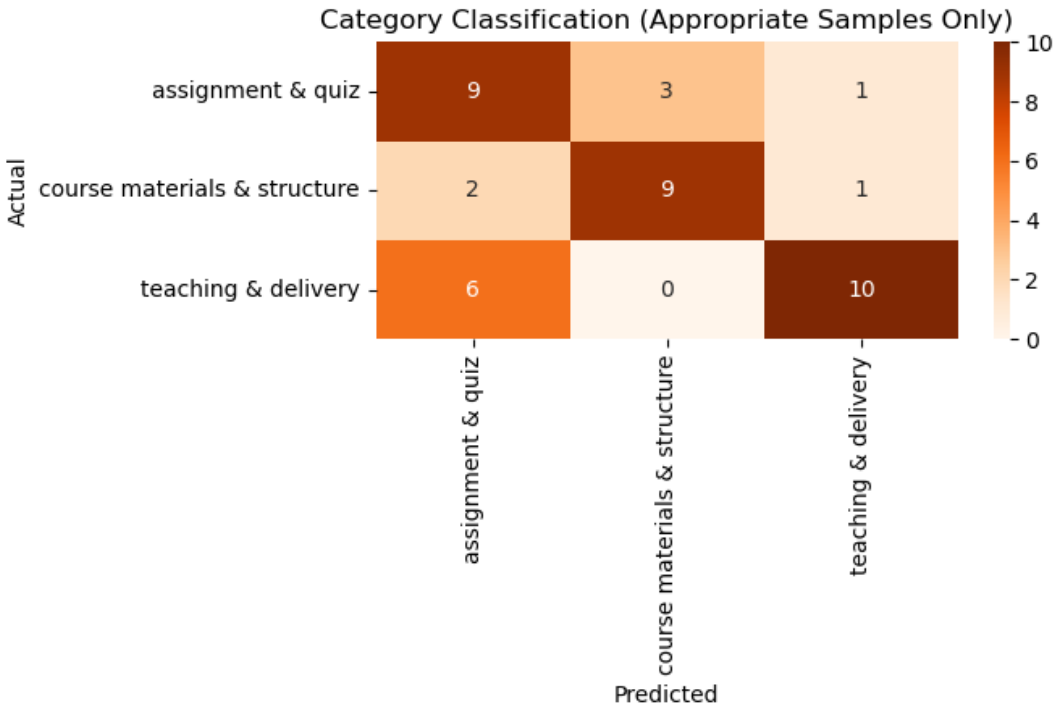
We tested the model to evaluate its stability and observed that while validation performance was strong, **test performance showed a noticeable drop.** Although accuracy decreased significantly, the macro F1-score remained moderately strong, indicating stable performance on new, unseen data.

This result reflects the fact that the test data was generated separately with a different prompt, indicating that while the training and validation data share structural similarities, the test data is more distinct. These findings suggest the need to re-train the model with more diverse training data to improve generalization.

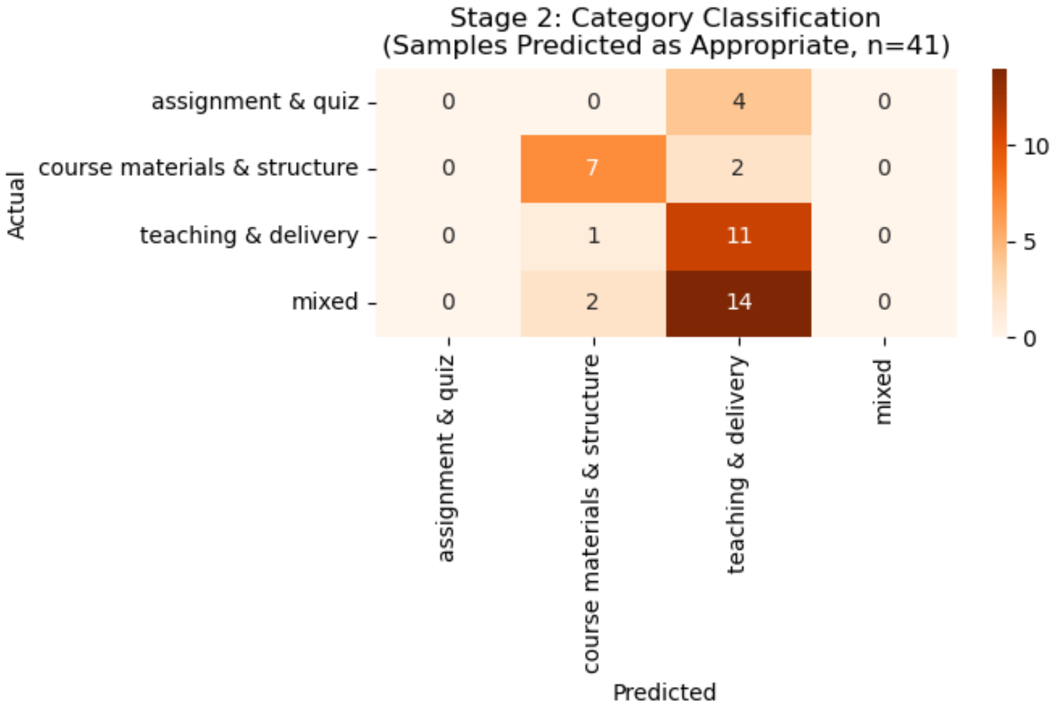
However, our results demonstrate that the model is powerful enough to perform the classification task effectively.

Visualisation – Categorisation Classifier

Multi-Task

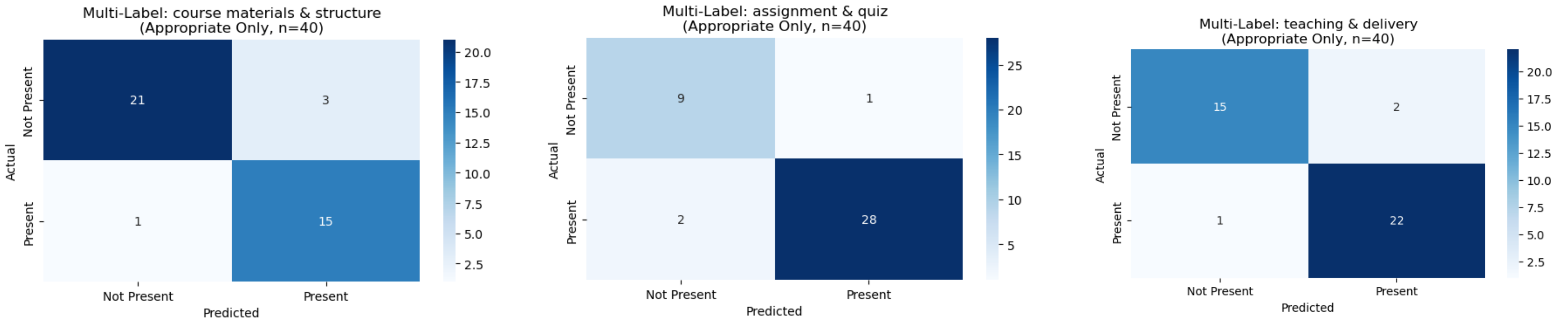


Cascaded Multi-Class



Visualisation – Categorisation Classifier

Cascaded Multi-Label



The categorisation classifier shows **better separation only in the multi-label approach**. For the single-label approach, we can see that using three labels in the multi-task structure still performs decently, with some misclassification across labels. However, the classifier performs poorly in the cascaded structure using four labels, with 'mixed' used to label responses that contain more than one category. This shows **that responses mentioning more than one category cause confusion for the model** that make it difficult for model to achieve high accuracy.

Testing with SELT Responses in 2024

Data Proportion

Response Length	Number of Response
More than 10 characters	[redacted]
Less than 10 characters	
Total	

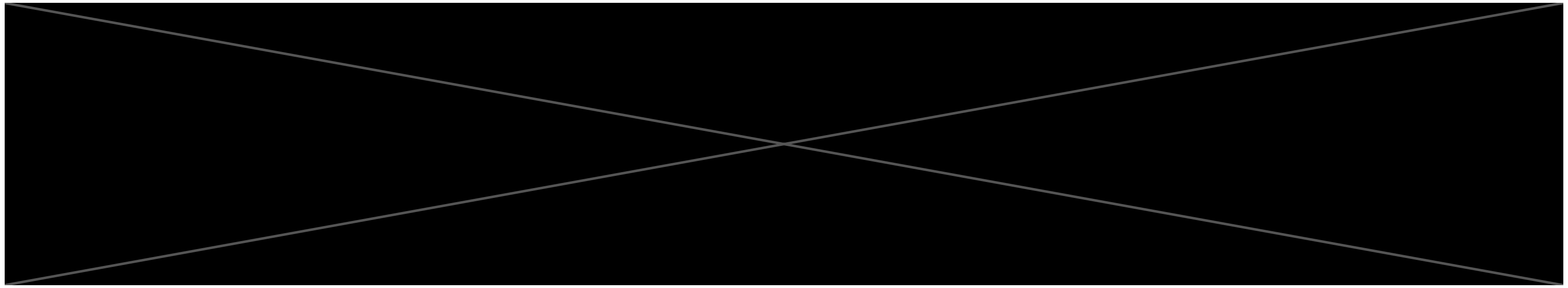
Results Using Cascaded Multi-Label

Model	Number of Response						
	Appropriateness		Content-based Categorization				
	Appropriate	Inappropriate	Course materials & structure	Teaching & delivery	Assignment & quiz	Multi-label	No label
DeBERTa	[redacted]						
RoBERTa							
Total							

Testing with SELT Responses in 2024

Key Insights – Appropriateness

- Response flagged as inappropriate are response that display negative sentiment or sensitive wording (rude, racism, and unprofessional) without providing constructive feedback for improvement.



- **Misclassification**:
    - From our review, we haven’t found any inappropriate responses that were misclassified as appropriate (false negatives).
    - Instead, most errors occur when appropriate responses are incorrectly flagged as inappropriate (false positives). This demonstrates that the model tends to **overpredict the inappropriate label.**

- Complaints directed at the course or teaching staff without actionable suggestions were also flagged as inappropriate, consistent with the training data. However, management may need to decide whether to continue treating these as inappropriate, since they may still provide useful signals about courses requiring attention.

Testing with SELT Responses in 2024

Key Insights – Appropriateness

- **Overpredict to inappropriate label**:
    - **Borderline response**: Many false positives fall in the probability range of 0.3–0.5 that reflects model uncertainty on borderline cases.

[redacted]

    - **Obvious misclassifications**: At the same time, some strongly positive responses are confidently mislabeled as inappropriate, which indicates the model has learned strong associations between specific words or structures to the inappropriate label.

[redacted]

- Overall, while false positives reduce the visibility of some appropriate responses, this behavior may be acceptable in the current context. Since we are focusing on **preventing inappropriate responses from not being identified.** However, this could be improved by retraining the model with a diverse set of responses.

Limitations – Synthetic Data

Data (Appropriateness):
In the synthetic data, inappropriate responses had strong negative sentiment, while appropriate response show positive sentiment. This creates straightforward separation between classes with very few “gray-area” examples that limits the model’s exposed to nuanced patterns and borderline cases.

Appropriate Responses

question	student_response	inappropriate_appropriate
What are the best aspects of this course?	the project-based learning approach was highly effective in developing real-world programming skills. working in teams of four students simulated industry conditions and taught valuable collaboration techniques. the course materials included current industry standards and best practices, with regular updates to reflect technological changes. the gitlab integration for version control was particularly useful, and the continuous integration pipelines helped students understand modern development workflows. assessment criteria were clearly communicated and consistently applied.	appropriate
What are the best aspects of this course?	the laboratory component was excellently designed with clear protocols and safety guidelines that were consistently enforced. students had access to modern equipment including spectrophotometers and centrifuges that are commonly used in research settings. the integration between theoretical concepts and practical work was seamless, with lab experiments directly supporting lecture material. however, some of the preparatory materials could be updated to reflect recent advances in biochemical techniques and instrumentation.	appropriate
What are the best aspects of this course?	the weekly problem sets were well-designed to reinforce lecture concepts but some of the more complex integration problems lacked sufficient worked examples in the course materials. students would benefit from additional practice problems with step-by-step solutions, particularly for topics like integration by parts and partial fractions. the online mathematics software was helpful for checking answers but occasionally had input format issues that caused frustration when submitting solutions.	appropriate

Inappropriate Responses

question	student_response	inappropriate_appropriate
Which aspects of this course need improvement?	the teaching assistant clearly has no idea what they're doing and shouldn't be allowed to teach.	inappropriate
Which aspects of this course need improvement?	these stupid professors don't know anything about teaching students properly and waste our time.	inappropriate
Which aspects of this course need improvement?	these asian students are always showing off and making everyone else look bad.	inappropriate
Which aspects of this course need improvement?	professor smith's teaching approach is completely ineffective and demonstrates a fundamental lack of understanding about how students learn mathematics. his lectures consist primarily of copying proofs from the textbook onto the blackboard without any explanation of the underlying intuition or practical applications. when students ask questions about concepts they're struggling with, he responds with condescension and impatience, often making comments that belittle students for not immediately grasping complex mathematical ideas. this creates a hostile learning environment where students are afraid to seek clarification, ultimately harming everyone's understanding of the material. his office hours are essentially useless because he treats every question as an interruption to his research work. the man clearly doesn't want to be teaching and has no business being in a classroom with undergraduate students. the assignments he creates are poorly designed, often containing errors that waste students' time and create confusion about correct solutions. when these errors are pointed out, he becomes defensive rather than acknowledging the mistakes and providing corrections. his grading is inconsistent and arbitrary, with identical solutions receiving different marks depending on his mood. the course content itself is important and fascinating, but his presentation makes it seem dry and irrelevant. students are suffering and learning less because of his complete inability to communicate mathematical concepts effectively. something needs to be done about this situation - either provide him with intensive pedagogical training or remove him from teaching responsibilities entirely. education is too important to be undermined by someone who clearly doesn't care about student learning.	inappropriate

Testing with SELT Responses in 2024

Single-Label – Content-Based Categorization

- The content-based classification seemed to perform better than we expected. From our review, we observed minimal misclassifications.

- This was surprising to us because we thought because of the limitations in the synthetic data, the model would struggle to generalize to real SELT data.

- It is also evident that the model performs well on shorter responses or those that clearly mention the category topic, thus performing more accurately in classify such responses.

Some of the 2024 SELT responses that were classified into one category

Question	Student Response Long	Course	pred_categories	pred_cat_prob
[redacted]			['assignment & quiz']	[0.03418615087866783, 0.03891736641526222, **0.9966819882392883**]
			['assignment & quiz']	[0.06989419460296631, 0.006062540225684643, **0.9898325204849243**]
			['assignment & quiz']	[0.019070260226726532, 0.015757344663143158, **0.9970058798789978**]
			['teaching & delivery']	[0.36802005767822266, **0.9957205653190613**, 0.009381748735904694]
			['teaching & delivery']	[0.010013390332460403, **0.9926026463508606**, 0.012474559247493744]
			['teaching & delivery']	[0.39943569898605347, **0.9748576283454895**, 0.004884300287812948]
			['course materials & structure']	[**0.975138783454895**, 0.1492191106081009, 0.0037436948623508215]
			['course materials & structure']	[**0.9963909983634949**, 0.04321946203708649, 0.007191849872469902]

Testing with SELT Responses in 2024

Multi-label – Content-Based Categorization

- The model also performs fairly well on multi-label categorization. However, we observed some misclassifications where responses that should belong to **a single category are predicted as multi-label**.
- This is likely due to the model **overfitting to specific keywords associated with each category.** These errors are more common in shorter responses, which tend to lack sufficient context or constructive feedback that makes it harder for the model to accurately capture the intended semantic meaning.
- Overall, similar to the appropriateness classification, this demonstrates that the model fundamentally performs well on classification tasks when trained with synthetic data, even though the data lacks variety. Performance could be further improved by retraining the model with actual SELT responses to introduce more diverse examples.

Some of the 2024 SELT responses with multiple categories that were correctly classified

Question	Student Response Long	Course	pred_categories	pred_cat_prob
[redacted]			['teaching & delivery', 'assignment & quiz']	[0.2981889247894287, **0.9969871640205383, 0.8332073092460632**]
			['teaching & delivery', 'assignment & quiz']	[0.23246723413467407, **0.9024419784545898, 0.9834743142127991**]
			['course materials & structure', 'assignment & quiz']	[**0.9833963513374329**, 0.07925458997488022, **0.8945600390434265**]

Some of the 2024 SELT responses with multiple categories that were misclassified

Question	Student Response Long	Course	pred_categories	pred_cat_prob
[redacted]			[~~'course materials & structure~~', 'teaching & delivery']	[**0.9431755542755127, 0.9842674136161804**, 0.0033240325283259153]
			[~~'course materials & structure'~~, 'teaching & delivery']	[**0.7135237455368042, 0.8775973320007324**, 0.0026275143027305603]
			['teaching & delivery', ~~'assignment & quiz']~~	[0.3075754642486572, **0.9932488203048706, 0.6848722696304321**]

Limitations – Synthetic Data

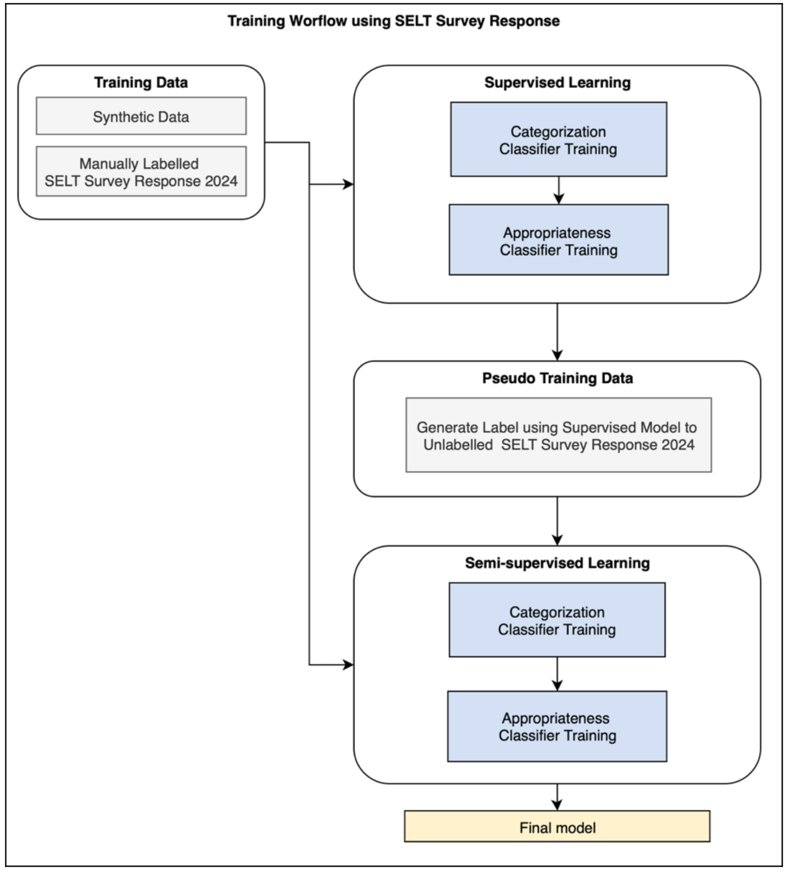
Data (Content-Based Categorisation):
Synthetic data generated by ClaudeAI contains repetitive structures with only slight changes in phrasing and course content. These minor variations do not provide sufficient diversity for the model that limits its ability to generalize in distinguishing between appropriateness and categorization labels. The model may overfit to this structured data and fail to generalize to text with different structures and nuances.

question	student_response	inappropriate_appropriate	all_labels
What are the best aspects of this course?	the integration of basic immunological principles with current research was excellent. the course materials included high-quality animations and interactive content that helped visualize complex immune processes. the teaching staff were active researchers who brought cutting-edge science into the classroom. the assignment tasks encouraged critical evaluation of scientific literature and development of research proposals.	appropriate	['course materials & structure', 'teaching & delivery', 'assignment & quiz']
What are the best aspects of this course?	the case study approach made the accounting principles more relevant and understandable. the course materials were well-organized with clear examples and practice problems. the teaching staff provided practical insights from their professional accounting experience. the assignment structure progressed logically from basic concepts to complex business scenarios. the software training component was valuable for developing professional skills.	appropriate	['course materials & structure', 'teaching & delivery', 'assignment & quiz']
What are the best aspects of this course?	the combination of theoretical rigor and practical applications made this course engaging. the course materials included excellent examples from various fields showing the relevance of probability theory. the teaching staff were patient in explaining difficult concepts and provided multiple approaches to problem-solving. the assignment feedback was detailed and helped identify areas for improvement.	appropriate	['course materials & structure', 'teaching & delivery', 'assignment & quiz']
What are the best aspects of this course?	the clinical placement component provided invaluable real-world experience working with mental health patients. the course materials addressed current mental health practices and evidence-based interventions. the teaching staff created a supportive learning environment for discussing challenging topics. the reflection assignments helped develop self-awareness and professional growth. however, the workload was quite intensive during clinical weeks.	appropriate	['course materials & structure', 'teaching & delivery', 'assignment & quiz']
What are the best aspects of this course?	the bloomberg terminal access was fantastic for learning real financial market analysis. the course materials included current market data and case studies from actual investment scenarios. the teaching staff had excellent industry connections and provided insights into professional investment practices. the group project simulated real portfolio management decisions and was highly engaging.	appropriate	['course materials & structure', 'teaching & delivery', 'assignment & quiz']

question	student_response	inappropriate_appropriate	all_labels
Which aspects of this course need improvement?	sql query practice is adequate but needs more complex real-world dataset examples rather than simple academic scenarios.	appropriate	['assignment & quiz']
Which aspects of this course need improvement?	penetration testing labs are exciting but need clearer boundaries and legal considerations explanation for students.	appropriate	['assignment & quiz']
Which aspects of this course need improvement?	operating system concepts are theoretical but need more hands-on administration experience with actual server environments.	appropriate	['assignment & quiz']
Which aspects of this course need improvement?	coding standards are important but project deadlines are unrealistic given the complexity of requirements and testing needed.	appropriate	['assignment & quiz']
Which aspects of this course need improvement?	problem-solving approaches are logical but need more step-by-step worked examples before independent problem attempts.	appropriate	['assignment & quiz']
Which aspects of this course need improvement?	mathematical proofs are challenging but office hours are insufficient for getting help with complex reasoning steps.	appropriate	['assignment & quiz']
Which aspects of this course need improvement?	formal logic systems are abstract but need more practical applications showing relevance to computer science problems.	appropriate	['assignment & quiz']

Training with SELT Response 2024

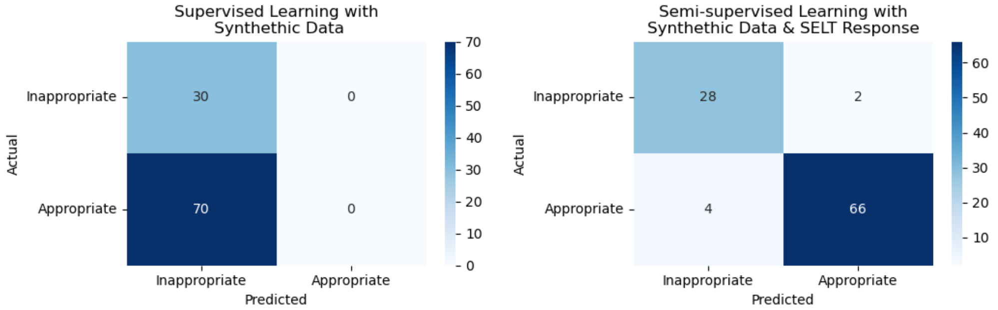
Semi-supervised Learning:
The base architecture of the model remained the same, but we trained it using 1000 rows of manually labeled SELT response 2024 data, combined with synthetic data. We then generated pseudo-labels using the first training iteration to enrich the training set and retrained the model.



Training with SELT Response 2024

Validation using 100 samples of labelled SELT response:

Appropriateness Classifier



Categorization Classifier

Model	Data	Accuracy	Macro F1-score
Supervised Learning	Synthetic Data	0.100	0.6880
Semi-supervised Learning	Synthetic Data & SELT Response	0.490	0.8173

Supervised Learning	Semi-supervised Learning
Extremely biased toward the inappropriate class, which makes it rarely detect appropriate comments correctly. This show the model's inability to perform well on actual data with higher complexity and structural variety.	**Shows significantly better performance in distinguishing between appropriate and inappropriate comments**
The model still achieves our critical goal, which is preventing inappropriate comments from slipping through.	Misses some inappropriate comments but performs better at distinguishing both classes.
Performs poorly in categorization. This further demonstrates the limitation of synthetic data that lacks variety.	**Shows significant improvement in categorization**, although improvement is still needed. This demonstrates the model's potential to achieve better generalization from using actual data with better labeling quality.
Struggles to capture textual nuance due to training only on synthetic data with similar structures.	Demonstrates the benefit of incorporating actual data during training and LLMs potential in capturing semantic understanding.

Common Pitfalls

Whilst experimenting, we noted several key takeaways that require close attention to avoid similar problems in future development:

1. **High learning rate:** This causes the model to struggle to converge, as it learns to quickly that leads to the loss to jump around instead of gradually decreasing. This can be seen in the validation loss, which fluctuates during the early epochs.

2. **Low epoch number:** Although a higher epoch increases computation, a low number of epochs prevents the model from fully learning and achieving optimal performance. This is reflected in both the training and validation loss, with the loss remaining high in both cases.

3. **Class weight:** Since our dataset has a slight imbalance between classes, applying class weights during training improves performance by giving more importance to the minority classes by increasing the penalty for mistakes on those classes. This helps the model learn them better and achieve a more balanced performance.

Summary and Future Improvement

In summary, our experiment revealed several important findings:

1. DeBERTa shows stronger performance in appropriateness, reflecting its theoretical advantage in capturing richer contextual understanding, whereas RoBERTa achieves slightly higher accuracy in categorization. This indicates a balance between the two models with each has strength in different task.
2. Using actual responses with diversity in structure and complexity in training helps the model capture better semantic understanding in distinguishing separation between class, specifically to our task, to achieve higher accuracy.

Therefore, with these findings we believe we can enhance our current model by implementing the following improvements:

<table>
<tr><th></th><th>Current</th><th>Potential Improvement</th></tr>
<tr><td rowspan="2">Training data</td><td>When reviewing the labels generated by our supervised model, we mainly focused on ensuring correct labelling for appropriateness and did not prioritise the categorisation labels.

This is likely the main cause of the worse performance in category classification.</td><td>Ensuring training data is correctly labelled to maintain high quality data.</td></tr>
<tr><td>Trained the model using only using the student response.</td><td>Perform context enrichment by concatenating the question to the survey response. This should provide additional context to help the model better differentiate across categories.</td></tr>
<tr><td rowspan="2">Model Architecture</td><td>Perform semi-supervised training using pseudo-labels with at least 90% confidence.</td><td>Apply a stricter threshold for pseudo-labels to avoid the risk of training on misclassified data.</td></tr>
<tr><td>Unfreezing 2 out of 12 transformer layers to fine-tune parameters for our classification task</td><td>Unfreeze more layers to allow the model to better adapt to our task.</td></tr>
</table>

Deployment and Integration

How can we deploy the model and integrate it with the current BI system?

Considerations: University-Specific Factors

Governance and Policy:

- Data Privacy
  - Student survey data often includes sensitive information. Deployment must comply with privacy laws and university data-handling policies.

Ethics:

- Bias and Fairness:
  - Ensure the model does not unfairly flag certain groups of students or writing styles as “inappropriate.”
- Transparency:
  - Staff should understand why a response was flagged or categorized.

Considerations: University-Specific Factors

Human Design:

- Oversight
  - The model should assist, not replace, human judgment. Staff may need the ability to review flagged responses before they are deleted.
- Feedback Loop
  - Provide a way for staff to correct the model's decisions, so performance can improve over time.

Usability:

- Integration with Existing Workflows:
  - The tool should fit smoothly into existing systems (Reporting, Data Warehouse, etc)
- Training and Support:
  - Provide documentation and training so staff understand what the model does, its limits, and how to use it responsibly.

Handover

Below are our documentation to ensure our project can be referred to for any plans to build upon in the future:

Code Documentation	We uploaded our training notebooks and trained models to a GitHub repository.
Detail Project Documentation	We prepared this final presentation that compiles our process, results, and insights from the project.